

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

64. Consider the following statements:

- I. If x and y are composite integers, so also is $x + y$.
II. If x and y are composite integers and $x > y$, then $x - y$ is also a composite integer.

III. If x and y are composite integers, so also in xy .

Which of the statement(s) given above is/are correct?

- (a) I, II and III (b) I and II
(c) Only III (d) None of these

65. Consider the following statements:

- I. The product of any three consecutive integers is divisible by 6.
II. Any integer can be expressed in one of the three forms $3k$, $3k + 1$, $3k + 2$, where k is an integer.

Which of the statement(s) given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

66. Consider the following statements:

- I. A natural number is divisible by 2, if its last digit is divisible by 2.
II. A natural number is divisible by 2, if its last digit is either zero or 2.
III. A natural number is divisible by 2, if its last digit is even.

Which of the statements given above are correct?

- (a) I and II (b) I and III (c) II and III (d) All of these

67. Consider the following statements for natural numbers a , b and c :

- I. If ' a ' is divisible by ' b ' and ' b ' is divisible by ' c ', then ' a ' must be divisible by ' c '.
II. If ' a ' is a factor of both ' b ' and ' c ', then ' a ' must be a factor of ' $b + c$ '.
III. If ' a ' is a factor of both ' b ' and ' c ', then ' a ' must be a factor of ' $b - c$ '.

Which of the statements given above are correct?

- (a) I, II and III (b) I and II
(c) II and III (d) None of these

68. Consider the following statements:

- I. Set of positive powers of 2 is closed under multiplication.
II. The set $\{1, 0, -1\}$ is closed under multiplication.
III. The number 35 has exactly four divisors.
IV. The set $\{1, 0, -1\}$ is closed under addition.

Which of the statements given above are correct?

- (a) I, II and III are true (b) Only III
(c) Only IV (d) All of these

PREVIOUS YEARS QUESTIONS

69. What number should be added to 231228 to make it exactly divisible by 33? **2012 I**
(a) 1 (b) 2 (c) 3 (d) 4

70. If a positive integer leaves remainder 28 when divided by 143, then what is the remainder obtained on dividing the same number by 13? **2012 I**
(a) 0 (b) 2 (c) 9 (d) 10

71. Which one of the following is neither prime number nor composite number? **2012 II**
(a) 1 (b) 2 (c) 3 (d) None of these

72. Which one of the following has least number of divisors? **2012 II**
(a) 88 (b) 91 (c) 96 (d) 99

73. How many numbers between -11 and 11 are multiples of 2 or 3? **2012 II**
(a) 11 (b) 14 (c) 15 (d) None of these

74. Consider the following statements
I. If n is a prime number greater than 5, then $n^4 - 1$ is divisible by 2400.
II. Every square number is of the form $5n$ or $(5n - 1)$ or $(5n + 1)$, where n is a whole number.

Which of the statement(s) given above is/are correct? **2012 II**

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

75. If N , $(N + 2)$ and $(N + 4)$ are prime numbers, then the number of possible solutions for N are **2013 I**
(a) 1 (b) 2 (c) 3 (d) None of these

76. The two-digit number, which when divided by sum of the digits and product of the digits, respectively leaves the same remainder the number is and the difference of quotients is one. **2013 I**
(a) 14 (b) 23 (c) 32 (d) 41

77. If x is positive even integer and y is negative odd integer, then x^y is **2013 I**
(a) odd integer (b) even integer
(c) rational number (d) None of these

78. Consider the following statements
I. There is a finite number of rational numbers between any two rational numbers.
II. There is an infinite number of rational numbers between any two rational numbers.
III. There is a finite number of irrational number between any two rational numbers.

Which of the statement(s) given above is/are correct? **2013 I**

- (a) Only I (b) Only II
(c) Only III (d) Both II and III

- 79.** If k is a positive integer, then every square integer is of the form **2013 II**
 (a) $4k$ (b) $4k$ or $4k + 3$
 (c) $4k + 1$ or $4k + 3$ (d) $4k$ or $4k + 1$
- 80.** If b is the largest square divisor of c and a^2 divides c , then which one of the following is correct (where, a , b and c are integers)? **2013 II**
 (a) b divides a (b) a does not divide b
 (c) a divides b (d) a and b are coprime
- 81.** Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ (where, k and m are integers), when **2013 II**
 (a) k is odd
 (b) k is even
 (c) k can be both odd and even
 (d) No such form is possible
- 82.** Consider the following statements
 I. 7710312401 is divisible by 11.
 II. 173 is a prime number.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 83.** Consider the following statements
 I. To obtain prime numbers less than 121, we are to reject all the multiples of 2, 3, 5 and 7.
 II. Every composite number less than 121 is divisible by a prime number less than 11.
 Which of the statement(s) given above is/are correct? **2013 II**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 84.** Consider the following statements
 I. No integer of the form $4k + 3$, where k an integer, can be expressed as the sum of two squares.
 II. Square of an odd integer can expressed in the form $8k + 1$, where k is an integer.
 Which of the statement(s) given above is/are correct? **2014 I**
 (a) Only I (b) Only II
 (c) Both I and II (d) Neither I nor II
- 85.** What is $26^2 + 97^2$ equal to? **2014 I**
 (a) $27^2 + 93^2$ (b) $34^2 + 93^2$ (c) $82^2 + 41^2$ (d) $79^2 + 62^2$
- 86.** If n is a whole number greater than 1, then $n^2(n^2 - 1)$ is always divisible by **2014 I**
 (a) 12 (b) 24 (c) 48 (d) 60
- 87.** What is the remainder when $(17^{23} + 23^{23} + 29^{23})$ is divided by 23? **2014 II**
 (a) 0 (b) 1 (c) 2 (d) 3
- 88.** What is the remainder when $(1235 \times 4523 \times 2451)$ is divided by 12? **2014 II**
 (a) 1 (b) 3 (c) 5 (d) 7
- 89.** What is the number of divisors of 360? **2014 II**
 (a) 12 (b) 18
 (c) 24 (d) None of these
- 90.** The multiplication of a three-digit number $XY5$ with digit Z yields $X215$. What is $X + Y + Z$ equal to? **2014 II**
 (a) 13 (b) 15 (c) 17 (d) 18
- 91.** How many pairs of X and Y are possible in the number $763X4Y2$, if the number is divisible by 9? **2014 II**
 (a) 8 (b) 9 (c) 10 (d) 11
- 92.** If $a_n = 3 - 4n$, then what is $a_1 + a_2 + a_3 + \dots + a_n$ equal to? **2014 II**
 (a) $-n(4n - 3)$ (b) $-n(2n - 1)$ (c) $-n^2$ (d) $-n(2n + 1)$
- 93.** Consider all those two-digit positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum? **2014 II**
 (a) 310 (b) 314 (c) 218 (d) 323
- 94.** p, q and r are prime numbers such that $p < q < r < 13$. In how many cases would $(p + q + r)$ also be a prime number? **2014 II**
 (a) 1 (b) 2 (c) 3 (d) None of these
- 95.** The digit in the units place of the product $81 \times 82 \times 83 \times 84 \times \dots \times 99$ is **2015 I**
 (a) 0 (b) 4 (c) 6 (d) 8
- 96.** What is the remainder obtained when $1421 \times 1423 \times 1425$ is divided by 12? **2015 I**
 (a) 1 (b) 2 (c) 3 (d) 4
- 97.** What is the remainder when 4^{96} is divided by 6? **2015 I**
 (a) 4 (b) 3
 (c) 2 (d) 1
- 98.** What is the maximum value of m , if the number $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$ is divisible by 5^m ? **2015 I**
 (a) 4 (b) 5
 (c) 6 (d) 7
- 99.** If $a - b = 4$ and $a^2 + b^2 = 40$, where a and b are positive integers, then $a^3 + b^6$ is equal to **2015 I**
 (a) 264 (b) 280
 (c) 300 (d) 324
- 100.** If n is a natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is **2015 I**
 (a) $x_1 + x_2 + x_3$ (b) $x_1 x_2 x_3$
 (c) $(x_1 + 1)(x_2 + 1)(x_3 + 1)$ (d) None of these

- 101.** If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y and z are natural numbers, then what is z equal to? **2015 I**
 (a) 1 (b) 2 (c) 3
 (d) Cannot be determined due to insufficient data
- 102.** A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was **2015 I**
 (a) 12 (b) 15 (c) 25 (d) 32
- 103.** The number of pairs (x, y) , where x, y are integers satisfying the equation $21x + 48y = 5$, is **2015 II**
 (a) Zero (b) One (c) Two (d) Infinity
- 104.** The largest natural number which divides every natural number of the form $(n^3 - n)(n - 2)$, where n is a natural number greater than 2, is **2015 II**
 (a) 6 (b) 12 (c) 24 (d) 48
- 105.** The digit in the units place of the resulting number of the expression $(234)^{100} + (234)^{101}$, is **2015 II**
 (a) 6 (b) 4 (c) 2 (d) 0
- 106.** A number when divided by 7 leaves a remainder 3 and the resulting quotient, when divided by 11 leaves a remainder 6. If the same number when divided by 11 leaves a remainder m and the resulting quotient when divided by 7 leaves a remainder n . What are the values of m and n , respectively? **2015 II**
 (a) 1 and 4 (b) 4 and 1 (c) 3 and 6 (d) 6 and 3
- 107.** The seven digit number $876p37q$ is divisible by 225. The values of p and q can be respectively **2015 II**
 (a) 9, 0 (b) 0, 0 (c) 0, 5 (d) 5, 9
- 108.** Let x and y be positive integers such that $x > y$. The expressions $3x + 2y$ and $2x + 3y$, when divided by 5 leave remainders 2 and 3, respectively. What is the remainder when $(x - y)$, is divided by 5? **2015 II**
 (a) 4 (b) 2 (c) 1 (d) 0
- 109.** The sum of first 47 terms of the series $\frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \frac{1}{4} - \frac{1}{5} + \frac{1}{6} + \frac{1}{4} + \frac{1}{5} - \frac{1}{6} - \dots$, is **2015 II**
 (a) 0 (b) $-\frac{1}{6}$ (c) $\frac{1}{6}$ (d) $\frac{9}{20}$
- 110.** The value of the expression $\frac{(243 + 647)^2 + (243 - 647)^2}{243 \times 243 + 647 \times 647}$ is equal to **2016 I**
 (a) 0 (b) 1 (c) 2 (d) 3
- 111.** What is the maximum value of m , if the number $N = 90 \times 42 \times 324 \times 55$ is divisible by 3^m ? **2016 I**
 (a) 8 (b) 7 (c) 6 (d) 5
- 112.** $7^{10} - 5^{10}$ is divisible by **2016 I**
 (a) 5 (b) 7 (c) 10 (d) 11
- 113.** If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of $a + b$ is (given that $a > 0$ and $b > 0$) **2016 I**
 (a) 7 (b) 9 (c) 11 (d) 13
- 114.** Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by $(x - m)$. Then, **2016 I**
 (a) R is a non-zero even integer
 (b) R is odd, if m is odd
 (c) $R = s^2$ for some integer s , if n is even
 (d) $R = t^3$ for some integer t , if 3 divides n
- 115.** If $\frac{61}{19} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$ where x, y and z are natural numbers, then what z is equal to? **2016 I**
 (a) 1 (b) 2 (c) 3 (d) 4
- 116.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct?
 I. $(y - x)$ can be an even integer.
 II. xy can be an even integer.
 III. $0.5(x + y)$ can be an even integer.
 Select the correct answer using the code given below **2016 I**
 (a) I and II (b) II and III
 (c) I and III (d) All of these
- 117.** Consider the following statements:
 I. Every natural number is a real number.
 II. Every real number is a rational number.
 III. Every integer is real number.
 IV. Every rational number is a real number.
 Which of the above statements are correct?
 (a) I, II and III (b) I, III and IV **2016 I**
 (c) II and III (d) III and IV
- 118.** Consider the following statements in respect of two different non-zero integers p and q .
 I. For $(p + q)$ to be less than $(p - q)$, q must be negative.
 II. For $(p + q)$ to be greater than $(p - q)$, both p and q must be positive.
 Which of the above statements is/are correct?
 (a) Only I (b) Only II **2016 I**
 (c) Both I and II (d) Neither I nor II

119. If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?

- I. $a - b < a - c$
 II. If $a < b$, then $\frac{a}{c} < \frac{b}{c}$
 III. $\frac{1}{b} < \frac{1}{c}$

Select the correct answer using the code given below

- (a) Only I (b) Only II
 (c) Only III (d) II and III

☑ 2016 I

120. If m and n are distinct natural numbers, then which of the following is/are integer/integers?

- I. $\frac{m}{n} + \frac{n}{m}$ II. $mn\left(\frac{m}{n} + \frac{n}{m}\right)(m^2 + n^2)^{-1}$
 III. $\frac{mn}{m^2 + n^2}$

Select the correct answer using the code given below

- (a) I and II (b) Only II
 (c) II and III (d) Only III

☑ 2016 (I)

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	
31		32		33		34		35		36		37		38		39		40	
41		42		43		44		45		46		47		48		49		50	
51		52		53		54		55		56		57		58		59		60	
61		62		63		64		65		66		67		68		69		70	
71		72		73		74		75		76		77		78		79		80	
81		82		83		84		85		86		87		88		89		90	
91		92		93		94		95		96		97		98		99		100	
101		102		103		104		105		106		107		108		109		110	
111		112		113		114		115		116		117		118		119		120	

HINTS AND SOLUTIONS

- (a) $(a + b)$ always represent a natural number $\forall a, b \in \mathbb{N}$.
- (d) As, $y = qx + r$, so $0 \leq r < x$.
- (a) As p is prime, so p/a or p/b .
- (c) Consider the given options
 - 161 is divisible by 7. Hence, 161 is not a prime number.
 - Since, $1 + 7 + 1 = 9$. Therefore, 171 is divisible by 3. Hence, 171 is not a prime number.
 - $173 < (14)^2$
173 is not divisible by any of the numbers 2, 3, 5, 7, 11, 13. Hence, 173 is a prime number.
 - 221 is divisible by 13. Hence, 221 is not a prime number.
- (c) Consider, $2, 4 \in \mathbb{N}$
So, $\sqrt{4} = 2$, a natural number and $\sqrt{2}$ = irrational number.

- (b) We know that, the product of a rational number and an irrational number is an irrational number.
- (c) If $x = 2$, then, $x, x + 1$ and $x + 3$ are all prime numbers.
- (c) $\because \text{Dividend} = D \times Q + R$
Given, $D = 5Q$ and $D = 2R$
When $R = 15, D = 2 \times 15 = 30$
 $\therefore Q = \frac{D}{5} = \frac{30}{5} = 6$
 $\therefore \text{Dividend} = 30 \times 6 + 15 = 195$
- (d) As, $4 + 2 + * + 8 = 14 + *$; $42 * 8$ is divisible by 9, if $14 + *$ divisible by 9.
So, $14 + * = 18$ (nearest multiple of 9)
 $\Rightarrow * = 18 - 14 = 4$
- (d) Here, $\frac{16}{7} \times \frac{16}{7} - \frac{*}{7} \times \frac{9}{7} + \frac{9}{7} \times \frac{9}{7} = 1$
Put $* = x$

$$\Rightarrow \frac{256}{49} - \frac{9x}{49} + \frac{81}{49} = 1$$

$$\Rightarrow 256 + 81 - 9x = 49$$

$$\Rightarrow 9x = 288 \Rightarrow x = 32$$

$$\Rightarrow 9x = 288 \Rightarrow x = 32$$

$$\begin{aligned} \text{11. (a) By BODMAS, } 10 \div 4 + 6 \times 4 \\ = 10 \div (4 + 6) \times 4 \\ = 10 \div 10 \times 4 = 1 \times 4 = 4 \end{aligned}$$

$$\begin{aligned} \text{12. (a) } \frac{a}{4} = \frac{b}{5} = \frac{c}{6} = k \text{ (say)} \\ \Rightarrow a = 4k, b = 5k \text{ and } c = 6k \\ \text{So, } \frac{a+b+c}{b} = \frac{4k+5k+6k}{5k} = \frac{15k}{5k} = 3 \end{aligned}$$

$$\begin{aligned} \text{13. (d) } [a - b + (b - a)] \\ - [2a - 2b + (b - 2a)] \\ \Rightarrow [a - b + b - a] - [2a - 2b + b - 2a] \\ \Rightarrow 0 - [2a - 2a - b] = 0 + b = b \end{aligned}$$

14. (b) $10^k = \frac{0.0003325}{3.325}$ (given)

$$= \frac{3.325 \times 10^{-4}}{3.325} = 10^{-4}$$

$$\Rightarrow 10^k = 10^{-4}$$

$$\therefore k = -4$$

15. (c) $\sqrt{0.04 \times 0.4 \times x} = 0.4 \times 0.04 \times \sqrt{y}$ (given)

$$\therefore \frac{\sqrt{x}}{\sqrt{y}} = \frac{0.4 \times 0.04}{\sqrt{0.04 \times 0.4}}$$

$$\frac{x}{y} = \frac{0.4 \times 0.4 \times 0.04 \times 0.04}{0.04 \times 0.4} = 0.016$$

(squaring both sides)

16. (b) Since, unit digit in 7^4 is 1.

$$\therefore 7^{168} = (7^4)^{42} \text{ give unit digit } 1.$$

$$\therefore 7^{170} = 7^{168} \times 7^2 \text{ gives the unit digit as } 1 \times 7 \times 7 = 9$$

17. (c) Unit place in $7^4 = 1$, so unit place in 7^{68} is 1.

$$\therefore \text{Unit place in } 7^{68} \times 7^3 = 3. \text{ Similarly, unit place in } 6^{59} \text{ is } 6 \text{ and unit place in } 3^4 \text{ is } 1, \text{ so unit place in } 3^{65} = 3^{64} \times 3 = 3.$$

$$\therefore \text{Unit place in } 7^{71} \times 6^{59} \times 3^{65} \text{ is the unit place of } 3 \times 6 \times 3 = 4$$

18. (b) As, n is divided by 4 the remainder is 3, so

$$n = 4q + 3, \text{ where } q \text{ is quotient.}$$

$$\Rightarrow 2n = 8q + 6$$

$$\Rightarrow 2n = (8k + 4) + 2 = 4(2k + 1) + 2$$

$$\text{So, if } 2n \text{ is divided by 4 the quotient is } 2k + 1 \text{ and remainder is } 2.$$

19. (c) Given, $\frac{1 - \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} - \left(\frac{1}{6} - \frac{1}{48}\right)} \div \frac{1 \times \frac{1}{6} - \frac{1}{48}}{\frac{1}{4} \times \left(\frac{1}{6} - \frac{1}{48}\right)}$

$$= x$$

$$\Rightarrow x = \frac{12 - 8 - 1}{48} \div \frac{1 - \frac{1}{24}}{\frac{1}{4} \times \left(\frac{8 - 1}{48}\right)}$$

$$\Rightarrow x = \frac{\frac{3}{5}}{\frac{48}{5} \div \frac{48}{7 \times 48}}$$

$$\Rightarrow x = \frac{3}{5} \div \frac{4}{7} \Rightarrow x = \frac{21}{20}$$

20. (b) Clearly, absolute value is defined by $|x| = -x$.

21. (a) $\frac{x-y}{x+y} = \frac{x/y-1}{x/y+1} = \frac{3/5-1}{3/5+1}$

$$= \frac{(3-5)/5}{(3+5)/5} = \frac{-2}{8} = -\frac{1}{4}$$

22. (a) As, $\left|\frac{7-x}{3}\right| < 2$

$$\Rightarrow \frac{7-x}{3} < 2 \text{ or } -\left(\frac{7-x}{3}\right) < 2$$

$$[\because |x| < a \Rightarrow x < a \text{ or } -x < a]$$

$$\Rightarrow 7-x < 6 \text{ or } \frac{x-7}{3} < 2$$

$$\Rightarrow -x < -1 \text{ or } x-7 < 6$$

$$\Rightarrow x > 1 \text{ or } x < 13$$

$$\Rightarrow 1 < x < 13$$

23. (d) A number is divisible by 25 when its last 2 digits are either zero or divisible by 25.

24. (a) $\frac{(2.48)^2 - (1.52)^2}{0.96}$

$$= \frac{(2.48 - 1.52)(2.48 + 1.52)}{0.96}$$

$$[\because a^2 - b^2 = (a-b)(a+b)]$$

$$= \frac{0.96 \times 4}{0.96} = 4$$

25. (d) Given, number is 222222.

$$\text{Here, sum of digits}$$

$$= 2 + 2 + 2 + 2 + 2 + 2 = 12,$$

$$\text{which is divisible by 3. So, given number is divisible by 3.}$$

$$\text{Now, sum of odd terms of digits - Sum of even terms of digits} = 6 - 6 = 0,$$

$$\text{it is divisible by 11.}$$

$$\text{Also, in a number if a digit is repeated six times, then the number is divisible by 7, 11 and 13.}$$

$$\text{Hence, the given number is divisible by 3, 7 and 11.}$$

26. (a) Required numbers are of the form of $12q - 2$

$$\text{i.e. } 10, 22, 34, 46, 58, 70, 82, 94$$

$$\therefore \text{Total sum} = 10 + 22 + 34 + 46 + 58 + 70 + 82 + 94 = 416$$

27. (d) The last digit in the expansion of $(2457)^{754}$ is equal to last digit of $(7)^{754}$.

$$(7)^{754} = (7^4)^{188} \times 7^2 = 1 \times 7^2$$

$$= (7)^2 = 49$$

$$\text{Hence, last digit is 9.}$$

28. (d) On taking option (d),

$$\text{The reverse digit of 451 is 154.}$$

$$\text{Now, } 154 + 297 = 451 \text{ is equal to the original number.}$$

29. (d) Let $a = 7p + 5$ and $b = 7q + 4$

$$\text{where, } p \text{ and } q \text{ are natural numbers.}$$

$$\therefore ab = (7p + 5)(7q + 4)$$

$$ab = 49pq + (4p + 5q)7 + 20$$

$$= 7(7pq + 4p + 5q) + 7 \times 2 + 6$$

$$\text{when } ab \text{ is divided by 7, we get the remainder 6.}$$

30. (b) If k is any even positive integer, then $k^2 + 2k$ is divisible by 8 but may not be divisible by 24.

$$\text{Let } k = 2m, m \in N, \text{ then}$$

$$k^2 + k \cdot 2 = 4m^2 + 4m = 4m(m + 1)$$

$$\text{which is divisible by 8.}$$

31. (c) Given number is 2784936.

$$\text{Sum of digits at odd places} = 25 \text{ and sum of digits at even places} = 14$$

$$\therefore \text{Difference} = 25 - 14 = 11$$

$$\text{So, number is divisible by 11.}$$

$$\text{Also last three digits of } 2784936 \text{ i.e. } 936 \text{ is divisible by 8.}$$

$$\text{Hence, } 2784936 \text{ is divisible by both 8 and 11 i.e. } 88.$$

32. (c) A number divisible by 88, if it is divisible by 8 and 11.

$$\text{In the given options } 9944 \text{ and } 8888 \text{ are divisible by 88. Hence, maximum number is } 9944.$$

33. (d) $\because \frac{1}{4} = 0.25$ and $\frac{3}{4} = 0.75$

$$\text{Only option (d) lies between } 0.25 \text{ and } 0.75. \text{ Since } \frac{252}{1000} = 0.252$$

$$\text{and } \frac{748}{1000} = 0.748$$

34. (b) Difference of sums of even and odd places digit of 1254934

$$= (1 + 5 + 9 + 4) - (2 + 4 + 3)$$

$$= 19 - 9 = 10$$

$$\text{This number will be divisible by 11, after adding } x, \text{ if } x = 1.$$

$$\text{Also, the sum of digits of } 1254934$$

$$= 1 + 2 + 5 + 4 + 9 + 3 + 4 = 28$$

$$1254934 \text{ will be divisible by 3, after adding } y, \text{ if } y = -1$$

35. (d) Last digit in the expansion of 3^{4798}

$$= \text{Last digit in the expansion of } (3^4)^{1199} \cdot 3^2$$

$$= \text{Last digit in the expansion of } 3^2 = 9$$

36. (b) The given number is divisible by 9, if sum of the digits is divisible by 9.

$$\text{Here, sum of digits}$$

$$= 8 + 4 + 7 + 0 + 5 + 2 + \otimes = 26 + \otimes$$

$$\text{If } \otimes = 1, \text{ then } 26 + 1 = 27 \text{ is divisible by 9.}$$

37. (c) The total number of three-digit numbers with unit digit 7 and divisible by 11 are 187, 297, 407, 517, 627, 737, 847, 957.

$$\therefore \text{Total numbers} = 8$$